



W A L A R K E L E C T R O N I C S

Fundrum Engineering · Christchurch, New Zealand

E4 EDUCATOR

v2.0

Enceladus Educator

Complete Embedded Systems Curriculum

ESP32 · PlatformIO · TFT Display · WiFi · MQTT · OTA · ESP-NOW · I2S Audio

16

Tutorials
Tier 1 · Tier 2

5

Projects
Tier 3 integrated

5

Toolkit files
Reusable across projects

espressif32@6.6.0 · TFT_eSPI@2.5.43 · PubSubClient@2.8 · ElegantOTA@2.2.9 · arduinoFFT@1.6.2

Walark Electronics · Fundrum Engineering · Christchurch, New Zealand · May 2026



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Document conventions

T01–T07 Tier 1 tutorials: platform fundamentals. No display or WiFi required.

T08–T16 Tier 2 tutorials: core skills. TFT display and WiFi introduced from T08.

P01–P05 Tier 3 projects: integrated applications. All Tier 1 + Tier 2 skills applied.

TK1–TK5 Toolkit files: reusable .h files that travel with you from T03 onwards into every project.